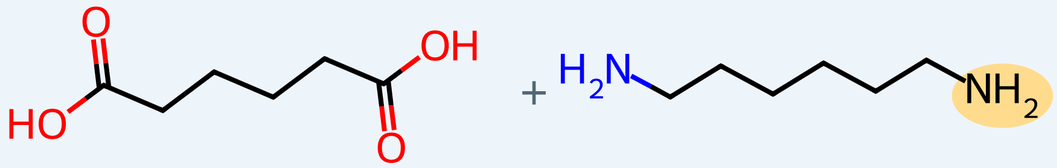
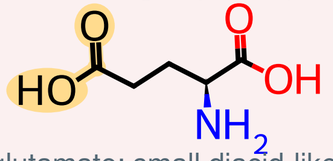
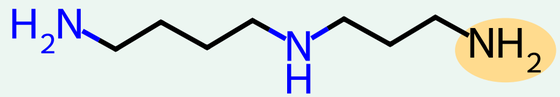
Chemical reasoning overrides substrate similarity

TARGET REACTION

adipic acid + hexamethylenediamine



free diacid + free diamine

CANDIDATE FAMILY (ATP-dependent amide ligase)	SUBSTRATE-SIMILARITY FIT	SUBSTRATE TOPOLOGY	DECISION
MptN · tetrahydromethanopterin: L-glutamate ligase IucA/RimK ATP-grasp fold (IPR041107)  glutamate: small diacid-like	0.83 ranked at the TOP by substrate structure + size	Glutamate is ligated onto tetrahydromethanopterin — a bulky cofactor, not a **free acid.** **No free-substrate pose.**	**STOP**
AsbA · spermidine–citrate ligase B. anthracis (A0A0F7RJ52) NIS synthetase, IucA/IucC (IPR022770)  free diamine (+ free citrate)	0.40 ranked LOWER by the same similarity metric	Free diamine + free diacid, condensed via acyl-adenylate **— same topology as target.**	**EXPAND**

Similarity put the carrier-tethered family on top (0.83 vs 0.40) — reasoning about substrate presentation reversed the call.

Recorded decision, canonical run 20260917_135330: fit = family_landscape.json; stop/expand = family_decisions.json. Prioritization, not a tested exclusion.